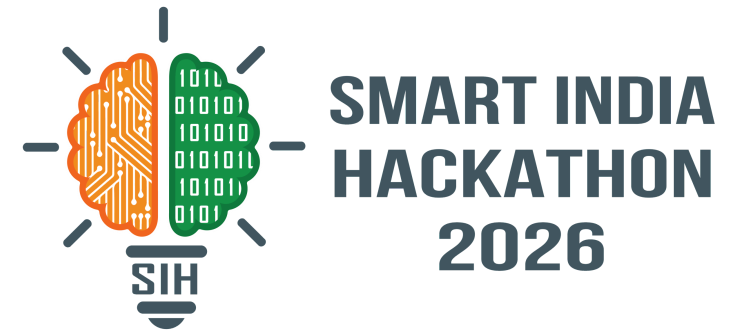


SMART INDIA HACKATHON 2026



Problem Statement ID –SIH26009

Problem Statement Title – Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls

Theme – Smart Automation

PS Category – Software

Team ID – SIH26-A0H-T506

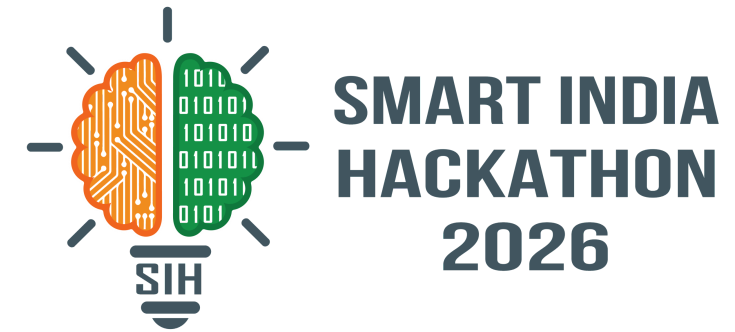
Team Name – SHANKR

A large, faint background graphic of a lightbulb with a circuit board pattern and binary code is centered on the page. The bulb is white with a grey base. The circuitry is orange and green, and the binary code is white. The bulb is surrounded by several grey hexagons of different sizes.

MANGAVISION AI
Space-assisted manganese intelligence



MANGAVISION AI – AI-POWERED PROSPECTIVITY MAPPING + PRODUCTION



Proposed Solution

A unified dashboard that fuses satellite observations, geological/borehole information and mine-operation history to rank manganese prospect zones, forecast production gaps and recommend corrective actions.

Detailed explanation

- Prospectivity engine: satellite spectral/terrain indicators + geology → confidence-ranked exploration zones.
- Shortfall engine: production targets + downtime + weather + blasting schedule → early risk forecast.
- Decision layer: converts the predicted gap into schedule, blasting and equipment-redeployment recommendations.

How it addresses the problem

Moves planning from reactive/manual estimation toward evidence-based, map-driven and predictive decisions; geological validation remains the final step before declaring a reserve.

Innovation and uniqueness

Two AI loops in one workflow: FIND the best exploration zones + PROTECT the production plan. Each output carries a confidence/risk score and an explainable reason.



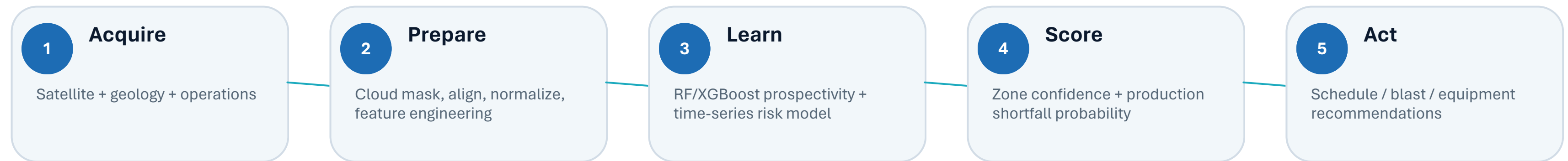
- **TECHNOLOGIES TO BE USED**

- Python • scikit-learn / XGBoost • PyTorch (optional) • GeoPandas / Rasterio • Google Earth Engine • FastAPI • PostgreSQL + PostGIS • Leaflet/Mapbox dashboard

- **DATA LAYERS**

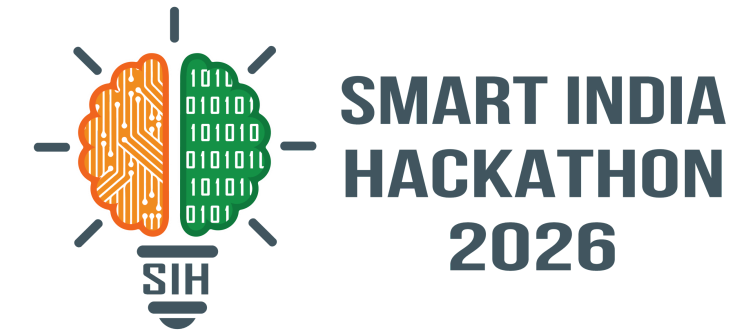
- Sentinel-2 / ASTER multispectral imagery • DEM / terrain • geological maps & borehole logs • historical production • equipment downtime • rainfall / soil moisture / LST • blasting schedule

- **METHODOLOGY / PROCESS**





FEASIBILITY AND VIABILITY



- **ANALYSIS OF FEASIBILITY**

- Open satellite data makes the geospatial MVP practical; operational/geological data can be demonstrated with anonymized or synthetic records until institutional access is available.
- The prototype can be built as a modular pipeline: data → models → GIS API → dashboard.

POTENTIAL CHALLENGES AND RISKS

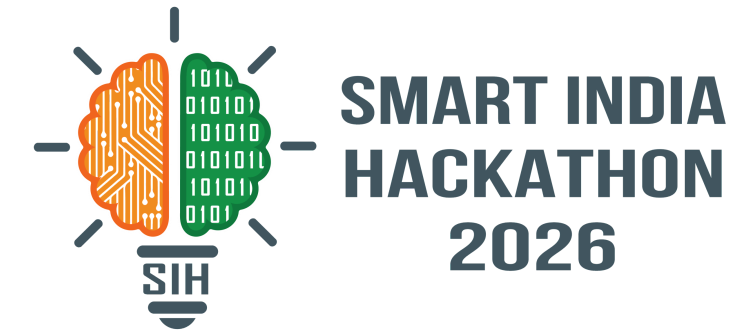
- Satellite signals are indirect and surface-focused; underground reserve claims cannot be validated from imagery alone.
- Mine production/equipment data may be proprietary or incomplete.
- Weather/clouds, missing logs and class imbalance can reduce model reliability.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

- Use satellite data for prospectivity screening, then require borehole/geological validation.
- Use confidence bands, explainable features and fallback rules when data is missing.
- Validate on known mineralized zones / historical production and report precision, recall, MAE and forecast lead time.



IMPACT AND BENEFITS



- **POTENTIAL IMPACT ON TARGET AUDIENCE**

- MOIL / mine planners: earlier warning of production gaps and clearer operational choices.
- Exploration teams: faster prioritization of high-potential zones for field investigation.
- Decision makers: one map + one risk dashboard for evidence-based planning.

BENEFITS — SOCIAL • ECONOMIC • ENVIRONMENTAL

ECONOMIC

Reduce avoidable production losses
Improve equipment utilization
Prioritize exploration spend

OPERATIONAL

Predict bottlenecks earlier
Support mine scheduling
Explain major risk drivers

ENVIRONMENTAL

Reduce unnecessary field surveying
Use targeted exploration
Support responsible mine planning

ZONE RANKING

High / Medium / Low

SHORTFALL RISK

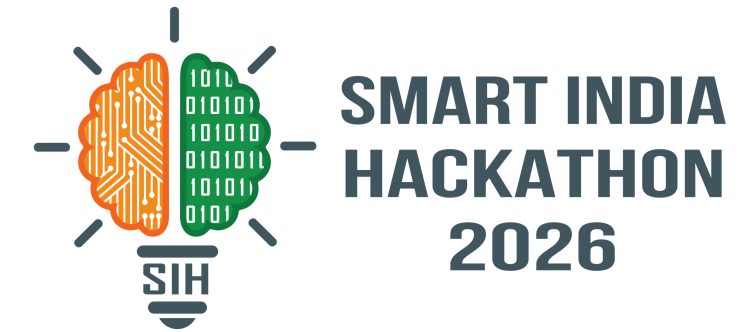
Probability + tonnes

RECOMMENDATION

Action + expected recovery



RESEARCH AND REFERENCES



- **RESEARCH AND REFERENCES**

1. Smart India Hackathon 2026 — Problem Statement SIH26009: Ministry of Steel / MOIL Ltd.
2. ESA — Copernicus Sentinel-2 mission: 13 multispectral bands; 10 m / 20 m / 60 m products; frequent revisit.
3. USGS — Scientific Investigations Map 3190: ASTER-based automated mineral-group mapping and mineral-resource assessment.
4. USGS — Scientific Investigations Map 3256: AVIRIS + ASTER remote-sensing mineral mapping.
5. Proposed validation: known mineralized zones + borehole logs + historical production; compare against baseline GIS/manual prioritization.

Key design principle

Satellite imagery is used for exploration screening and spatial indicators—not as a substitute for drilling/geological confirmation of subsurface reserves.

Reference links

- ESA Sentinel-2: esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-2
- USGS SIM 3190: usgs.gov/publications/sim3190
- USGS SIM 3256: usgs.gov/publications/sim3256